

$$9) \begin{bmatrix} 3 & -3 \\ 6 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 6 & 1 \\ 6 & -5 & 4 \end{bmatrix}$$

$$10) \begin{bmatrix} 3 & 1 \\ -3 & -4 \end{bmatrix} \cdot \begin{bmatrix} 1 & -3 \\ -4 & -1 \end{bmatrix}$$

$$11) \begin{bmatrix} 3 & 2 \\ 2 & 1 \\ 3 & 4 \\ -1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 2 & -6 \end{bmatrix} \cdot \begin{bmatrix} -2 & 6 \\ 0 & 0 \end{bmatrix}$$

$$12) \begin{bmatrix} 4 & -2 \\ -3 & 6 \end{bmatrix} \cdot \left(\begin{bmatrix} 4 & -5 & 4 & 0 \\ 6 & 3 & 0 & -3 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ -6 & -1 \end{bmatrix} \right)$$

$$13) \begin{bmatrix} 5ab & -1 & 0 \\ ba & 5b & -1 \end{bmatrix} \cdot \begin{bmatrix} b \\ 4 \\ 2b \end{bmatrix}$$

$$14) \begin{bmatrix} 6u & 2u & u^2 \\ uv & v^2 & -2v^2 \end{bmatrix} \cdot \begin{bmatrix} -3u & 1 \\ 6u & 3v \\ -6u & v \end{bmatrix}$$

Critical thinking questions:

15) Give an example of a matrix X that would make the expression AX defined where A is a 5×7 matrix.

16) How many multiplications of two numbers would be required to multiply a 3×5 matrix by a 5×10 matrix?